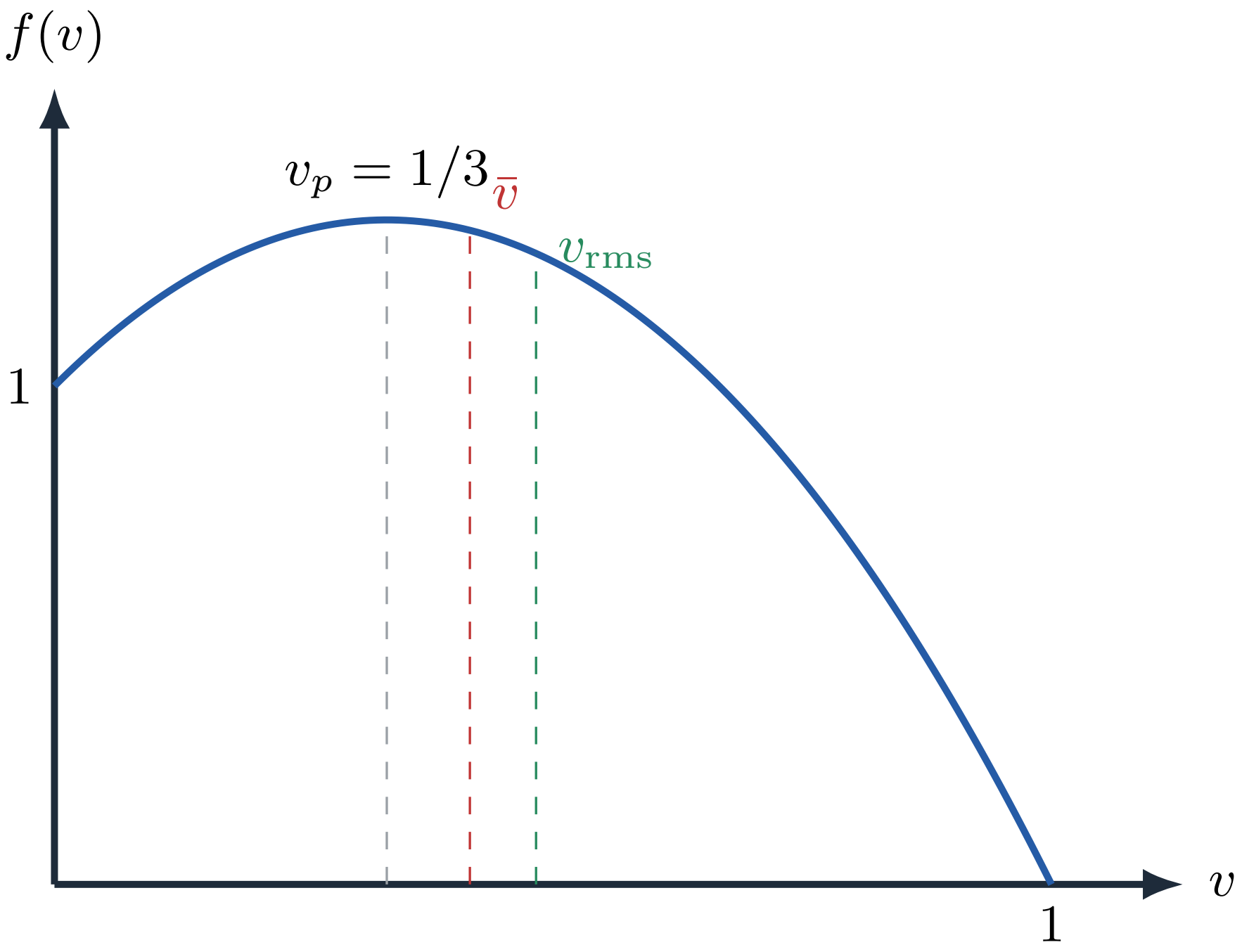

$$\int_0^1 f(v)\mathrm{d}v = 1$$